LLM Statements

For this project we used LLM for the following parts:
- For inspiration on which ICT-systems were needed for the third business opportunity.

Introduction

This paper was written by Max Harmsen (2335778), Matt ter Steege (9932003) and Marc Jan van Veldhuizen (1263285), all part of group 1-4. The paper will discuss business intelligence opportunities of IKEA based on the business processes which where modeled in the previous assignment (Harmsen et al., 2026). First, an ER diagram will be introduced that models the data model needed for the process of sourcing sustainable materials. The model is based on the to-be BPMN model of assignment 2 and the indicators mentioned in assignment 2 (Harmsen et al., 2026). Following the data model, 3 business intelligence opportunities will be addressed. These opportunities showcase what parts of the organization can be improved and how they could be improved. Lastly, the opportunities will be evaluated by looking what ICT systems are needed to make use of the opportunities.

1 | ER diagram of IKEA's data warehouse

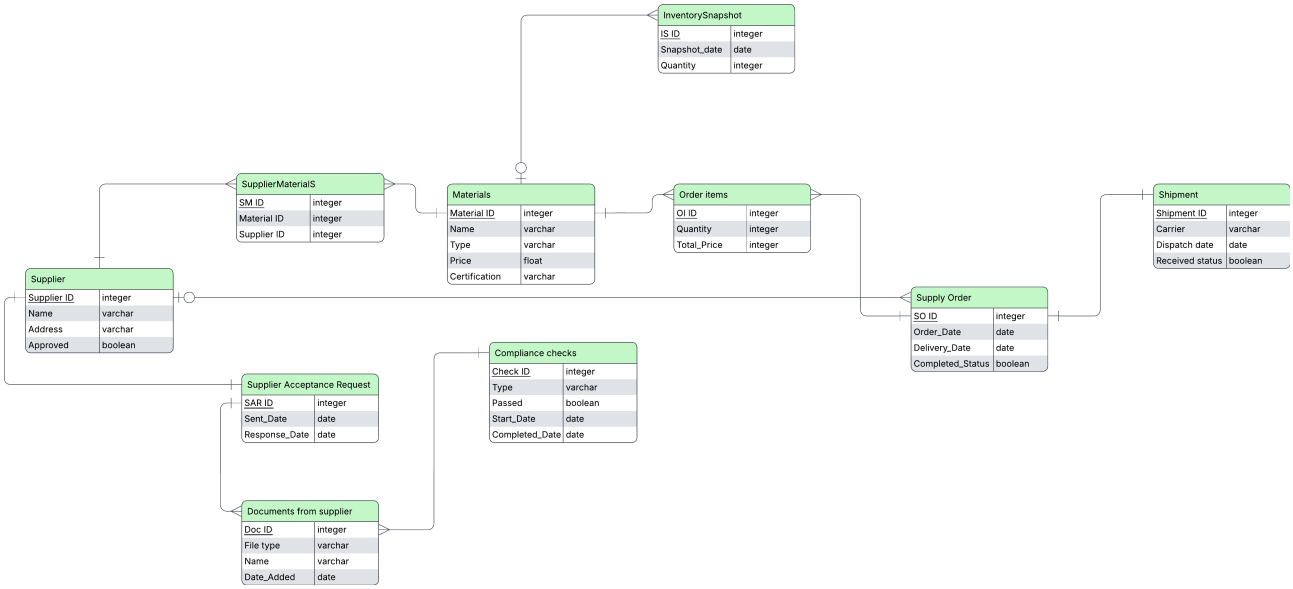

Figure 1: ER diagram representing all entities that are involved in the process of sourcing sustainable materials

Figure 1 shows the ER-diagram (according to the Crow's foot notation) that represents the process of sourcing sustainable materials. All the processes that are supported by this data model are shown in the to-be process found in our third assignment (Harmsen et al., 2026). In the diagram, every entity has one underlined attribute. This represents the primary key of that entity, meaning that attribute will be the unique identifier of a certain record of that entity.

Indicators

The dataset should at least contain enough information to calculate the indicators mentioned in the third assignment (Harmsen et al., 2026).

Indicator 1

The first of those indicators was dividing the total amount of requested suppliers by the total number of accepted suppliers. The data model supports this indicator by saving an *Approved* attribute which is a boolean. This means that every supplier that was every requested is saved in the **Supplier** table. After going through the

acceptance process, the supplier is either rejected or accepted. That result will then be represented in the *Approved* attribute as true or false. Since rejected suppliers are also saved in this table, one can use a simple

```
SELECT COUNT(*)
FROM Supplier
```

which will count all rows of the supplier table and result in the total number of requested suppliers. After that, the following script can be executed to evaluate how many suppliers where approved.

```
SELECT COUNT(*)
WHERE Approved = TRUE
FROM Supplier
```

Indicator 2

The second indicator states how long the average compliance check takes per new supplier. The data model makes this possible by saving *Start Date* and *Completed Date*. The difference between these 2 dates will result in the time that the check took. Since the type of check is also saved, and the check is linked to a certain supplier request, you know when all 4 checks for a certain supplier are finished.

Indicator 3

The last indicator that was described was the turn over rate. To calculate the turn over rate one should divide the total amount of materials used in one year by the average amount of stock for that material. To make sure this indicator could also be used, the data model includes the **InventorySnapshot** table. Using a certain frequency that is chosen by IKEA, this table will include the quantity of a certain material on that point. The frequency could be every day or week, but it could also be every month if the throughput of materials is not that high. Since each record within the **InventorySnapshot** contains a *Snapshot date*, the indicator can be calculated after each year has finished. The first and last snapshot of the year together with all the delivery quantities can be used to calculate the total use of that material in a year. The equation goes as followed:

$$Usage = FirstSnapshot + \sum Deliveries - LastSnapshot$$

$$AverageInventory = \frac{\sum Snapshots}{NumberOfSnapshots}$$

First and last mean of the certain year your the turnover of. Snapshot references to the quantity of said snapshot.

Then the final turnover rate would become.

$$TurnoverRate = \frac{Usage}{AverageInventory}$$

After that, one can calculate the average amount of stock by adding all the quantities of the snapshots and dividing that number by the total number of entries.

Choices

Even though most parts of our model are quite logical and strait forward, there are a few choices that need some more explanation. The first being the relationship between **Supply Order** and **Shipment**. For the simplicity of the model, there was a 1 to 1 relation implemented. This means that an order gets always sent in its entirety and not in multiple shipments. Even though this simplifies our data model, this choice could be argued well, since an order is always placed at 1 supplier. It therefore is not illogical at all that all the materials that that supplier supplies are sent in once to prevent high shipping costs. The second choice that has to be explained is the relation between **Supplier** and **Supplier Acceptance Request**. This 1 to 1 relation was chosen because before a supplier can be added to the **Supplier** table, the supplier already should have been requested. In addition, a supplier can only be requested once, since a request will always result in either a approved status or a rejected status. The rest of the entities and relationships do not need further explanation since the reason they are in place are trivial.

2| Business intelligence opportunities

Opportunity 1 – Predictive Inventory Replenishment

A company as large as IKEA has enormous amounts of materials to track. Managing this manually or with static thresholds quickly becomes inefficient, especially when demand fluctuates over time.

That is the exact reason why an automatic ordering system should be set into place. As seen in Figure 1, periodic snapshots (**InventorySnapshot**) are taken of material stock levels. These snapshots provide a historical record of how inventory of a certain material evolves over time.

This data can be aggregated to compute material consumption rates by comparing stock levels between consecutive snapshots. When combined with transactional data such as sales or production usage, it becomes possible to derive accurate demand patterns for each material.

Based on this aggregated dataset, predictive models can be trained to forecast future demand. For example, time-series forecasting techniques can be applied to identify trends, seasonality, and recurring peaks in material usage. This allows the system to estimate how quickly current stock levels will be depleted.

In addition, supplier data such as lead times can be integrated into the model. By combining predicted demand with lead times, the system can determine the optimal moment to reorder materials and the appropriate order quantities.

The result is a business intelligence solution in the form of a decision-support dashboard and automated triggers, which:

- Display predicted stock depletion dates per material
- Highlight high-risk items that are likely to go out of stock
- Recommend reorder quantities based on expected demand

This system would primarily be used by supply chain managers and procurement teams to support operational decision-making. It directly contributes to improving operational efficiency by reducing stockouts, minimizing excess inventory, and ensuring continuity in production and sales processes.

Opportunity 2 – Supply Chain Traceability Analytics

One of the recurring issues in large-scale supply chains is the lack of end-to-end visibility. For a company operating at IKEA's scale, materials and products move through multiple stages, including suppliers, distribution centers, and retail locations. Without proper traceability, it becomes difficult to identify where delays, defects, or inefficiencies originate.

To address improvement goal IG2.4 (improve traceability), the organisation can exploit data already collected across its supply chain systems. This includes:

- Shipment records between suppliers, warehouses, and stores
- Timestamps of material movements
- Supplier identifiers and batch numbers
- Inventory updates at each location

By aggregating these datasets, a complete timeline of each product or material batch can be reconstructed. This effectively creates a traceability chain, where every movement and transformation step is recorded and linked. Once this data is structured, business intelligence techniques can be applied to compute key indicators such as:

- Average transit time between supply chain stages
- Frequency and location of delays
- Defect rates per supplier or batch

More advanced analysis can be performed using anomaly detection to identify irregular patterns, such as unexpected delays or missing tracking events. This allows the organisation to quickly detect disruptions or inconsistencies in the supply chain. The resulting BI solution would take the form of a traceability dashboard, where users can:

- Track the full life cycle of a product or material batch
- Identify bottlenecks in the supply chain
- Compare supplier performance based on reliability and quality
- Track back where faulty batches originated from and where they are now

This system would primarily be used by logistics managers and quality control teams. It directly contributes to improved traceability by increasing transparency, enabling faster issue resolution, and supporting more informed supplier and logistics decisions.

Opportunity 3 – Take-Back Program Optimization and Forecasting

Expanding take-back programs introduces a different kind of complexity compared to traditional forward logistics. Returned products vary in condition, timing, and location, making it difficult to plan processing capacity and resource allocation effectively.

To support improvement goal IG3.3 (expand on take-back programs), the organisation can exploit data generated through return flows and recycling initiatives. Relevant data sources include:
- Records of returned products (type, quantity, condition)
- Return locations and timestamps
- Processing outcomes (reused, recycled, discarded)
- Customer participation rates per region

By aggregating this data over time and across locations, patterns in return behavior can be identified. For example, certain product categories may have consistently higher return rates, or specific regions may show stronger participation in take-back initiatives.

Using this aggregated dataset, predictive analytics can be applied to forecast future return volumes. This can be done by analyzing historical return trends and identifying seasonal effects or campaign-driven spikes.

These forecasts enable the organisation to:
- Anticipate the volume of returned goods per location
- Allocate processing capacity more efficiently
- Optimize logistics for reverse supply chains

In addition, clustering techniques can be used to segment products or regions based on return characteristics, allowing for more targeted improvement strategies.

The BI solution would be implemented as a dashboard that:
- Visualizes return volumes and trends over time
- Highlights high-performing regions in terms of participation
- Provides forecasts for incoming returns

This system would be used by sustainability managers and logistics planners. It supports the expansion of take-back programs by making them more predictable, scalable, and cost-efficient, while also improving overall resource recovery.

3 | ICT-systems needed

The opportunities mentioned above will need support of ICT-systems. These systems will be discussed in this part of the paper. This will be done by addressing each opportunity and give an explanation about which ICT system is needed and the reason this system is needed. The systems that will be evaluated are addressed in the following order: Predictive Inventory Replenishment, Supply Chain Traceability Analytics and Take-Back Program Optimization and Forecasting.

Predictive Inventory Replenishment

To implement an predictive inventory restocking system there are a few ICT-systems needed. The first thing needed is a Business Intelligence system which will take all the historical **InventorySnapshots** and calculates stock changes between the snapshots. The system will also need to contain a predictive element so the trends in sales can be recognized. Those trends predictions can be made about which quantity of products are needed. The system should also include decision supporting, so the managers can make informed decisions based on the data provided to them through historical records. Their should also be a Knowledge management system this will turn the raw data from the BI system into a report, this also will take the standard procedures and rules about orders in consideration. For the KM system there is also a need for automated triggers so that either recommendations or direct orders can be made.

Supply Chain Traceability Analytics

In this case their is the need for a Business Intelligence system which will execute a multidimensional data analysis so that the data can be seen in small parts. This system should also be able detect if their is a unusual pattern in the supply chain. The system can implement the detection by using sequences mining which will detect any disruption of the supply chain and shows where it happens. ("Understanding AI Business Intelligence: A Comprehensive Guide", n.d.) Secondly, a knowledge management system is needed to give the disruption found by the business intelligence system context. This means providing answers like where and

why happens the disruption. The knowledge management system will be the crucial link between the business intelligence systems and other peripheral systems used within the organizations. Those peripheral systems can be emails, but also order management systems.

Take-Back Program Optimization and Forecasting.

This opportunity needs a Business Intelligence system which will need a few key components. The first being a Data Warehouse to keep a record of the data from all different inputs. Secondly, there is a need Data Cleansing. This is the process where the inputted data is analyzed, compared to the desired data and when rewritten to follow the wanted pattern ("What Is Data Cleaning?", 2024). For example, one of the employees wrote the date in the MM-DD-YYYY format while the system requires DD-MM-YYYY. The data cleansing is needed because the following system will need the data to be as structured as possible to get the best results. The just mentioned follow up system will by a data mining system. This system focuses on forecasting needed capacity and then distributing this forecast to support systems which directly support managers.

4 | Conclusion

To conclude, 10 entities where needed to create a complete data model needed for the sourcing of sustainable materials. These entities where modeled in an ER-diagram using the Crow's foot notation. Following the ER diagram, 3 opportunities where concluded. These opportunities where

- Predictive Inventory Replenishment
- Supply Chain Traceability Analytics
- Take-Back Program Optimization and Forecasting

Each opportunity describes a way where IKEA could improve their business processes using business intelligence. Business intelligence processes need ICT systems to function. These ICT-systems where evaluated in the last part of the paper.

References

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